

CSC G0040

The Cyberattack Lifecycle: Threat Intelligence, Digital Forensics, and Incident Response

Final Project

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Digital Forensic Analysis of Insider Data Leakage on a Company-Issued iPad

Abstract

Insider threats pose a significant and persistent risk to organizations, particularly in environments where employees are granted legitimate access to sensitive corporate information as part of their job responsibilities. Unlike external attacks, insider incidents are often more difficult to detect because the actions may appear normal within authorized system boundaries. This paper presents a digital forensic investigation of a simulated insider data-leak incident involving a company-issued iPad used by an employee with approved access to confidential business files. The scenario focuses on how sensitive corporate documents can be misused and transferred to personal cloud-storage services without proper authorization, highlighting a common risk faced by modern organizations that rely on mobile devices for daily operations.

To examine this incident, a full unencrypted iTunes backup of the iPad was acquired in a forensically sound manner and analyzed using the Autopsy digital forensics platform within a virtualized environment. The analysis concentrated on identifying user activity artifacts, file metadata, application data, and timeline information that could

indicate potential data exfiltration. The investigation demonstrates how logical iOS backups can preserve valuable forensic evidence, such as file remnants and application usage data, which can support internal investigations of insider threats. However, the findings also reveal important limitations of iTunes backups, including the absence of detailed file access logs, cloud transaction records, and certain system-level artifacts. Overall, this study highlights both the strengths and constraints of iOS logical backups and underscores the need for complementary forensic techniques when investigating insider data-leak incidents on mobile devices.

Introduction

Mobile devices such as smartphones and tablets have become essential tools in modern organizations, enabling employees to access corporate documents, communicate with colleagues, and perform daily work tasks with greater efficiency and flexibility. Organizations increasingly rely on mobile platforms to support remote work, collaboration, and real-time decision-making. However, while these devices significantly enhance productivity, they also introduce substantial security risks, particularly when they store or provide access to sensitive corporate data outside of traditional enterprise network boundaries.

One of the most challenging security risks associated with mobile devices is insider data leakage, where authorized users intentionally or unintentionally misuse their legitimate access to expose confidential information. Unlike external cyberattacks, insider incidents are difficult to detect because the activities often appear normal and are performed using valid credentials and approved applications. This makes traditional security controls and intrusion detection systems less effective in identifying malicious behavior. As a result, organizations may remain unaware of insider misuse until significant data loss has already occurred.

This project focuses on understanding how digital forensic techniques can be applied to investigate insider misuse on a company-issued iPad. By simulating a data-leak scenario, the investigation demonstrates how forensic analysis of iOS devices can uncover evidence of unauthorized data handling, such as file access artifacts, application usage records, and metadata preserved in device backups. The project also explores the limitations of mobile forensic investigations, particularly when relying on logical backups, and emphasizes the importance of combining technical analysis with proper investigative methodology to support insider-threat detection and response.

Background and Related Work

Digital forensics is the systematic process of identifying, preserving, analyzing, and presenting digital evidence in a structured and legally defensible manner. It plays a critical role in modern investigations by enabling examiners to reconstruct events, determine user actions, and support legal or organizational decision-making. Mobile device forensics is a specialized subfield within digital forensics that focuses on the examination of smartphones and tablets, which present unique challenges compared to traditional computers. These challenges include strong encryption mechanisms, proprietary operating systems, limited file-system access, and application sandboxing, all of which can significantly restrict the availability of forensic artifacts.

Apple's iOS operating system is specifically designed with strong security and privacy controls to protect user data and prevent unauthorized access. Features such as file-based encryption, secure enclaves, and strict application isolation enhance device security but simultaneously complicate forensic investigations. As a result, investigators often face limitations when attempting to access system-level logs, detailed file-access records, and cloud synchronization activity. Direct physical acquisition of iOS devices is frequently not feasible without advanced tools or elevated privileges, making logical acquisition methods more commonly used in practice.

Previous research has demonstrated that iTunes backups, particularly unencrypted backups, can still provide valuable logical forensic artifacts despite these limitations. Such backups may contain application data, file metadata, configuration files, and usage artifacts that help reconstruct user activity and support investigative conclusions. However, many low-level system events, including detailed file access logs and cloud transaction records, remain inaccessible through iTunes backups alone. These findings highlight both the usefulness and the constraints of logical iOS backups and emphasize the importance of understanding their scope when conducting mobile forensic investigations.

Incident Scenario

The simulated incident examined in this project involves an employee who is issued an iPad by an organization for official work-related purposes. The device is provisioned and managed to allow access to confidential corporate documents that are required for the employee's daily responsibilities. These documents are accessed through the iOS Files application and various integrated cloud-storage services, which are commonly used in modern enterprise environments to support collaboration, remote work, and document synchronization across multiple devices. The organization operates under the assumption that the employee will comply with internal data-handling policies and use the company-issued device in a responsible and secure manner.

In this scenario, the employee intentionally misuses their legitimate access to sensitive corporate information. The user opens and reviews confidential files stored on the iPad, copies them to alternative directories, deletes original files to reduce visibility, and uploads the data to personal cloud-storage accounts such as Google Drive and iCloud Drive without authorization. These actions are performed using built-in iOS applications and trusted cloud services, allowing the activity to blend in with normal device usage. As a result, traditional security monitoring tools may fail to identify the behavior as malicious in real time. This form of misuse illustrates a common insider data-exfiltration technique, where trusted users exploit their authorized privileges to bypass organizational controls and transfer sensitive information beyond corporate boundaries.

The simulated incident highlights the complexity of insider-threat investigations involving mobile devices. Because the employee's actions do not involve external hacking tools or malware, the activity leaves limited forensic traces at the system level. Instead, investigators must rely on logical artifacts such as file metadata, application data, and backup contents to reconstruct the sequence of events. This scenario demonstrates how insider data leakage can occur quietly and emphasizes the importance of digital forensic analysis in identifying, understanding, and documenting unauthorized data handling on company-issued mobile devices.

Methodology

The investigation followed a structured and forensically sound methodology designed to simulate and analyze insider misuse on a company-issued iPad. First, suspicious user activities were deliberately performed on the device to replicate a realistic insider data-leak scenario. These activities included opening confidential documents, copying and deleting files, and uploading sensitive data to personal cloud-storage services using standard iOS applications. This step ensured that the simulated incident closely reflected common insider behavior observed in real-world cases.

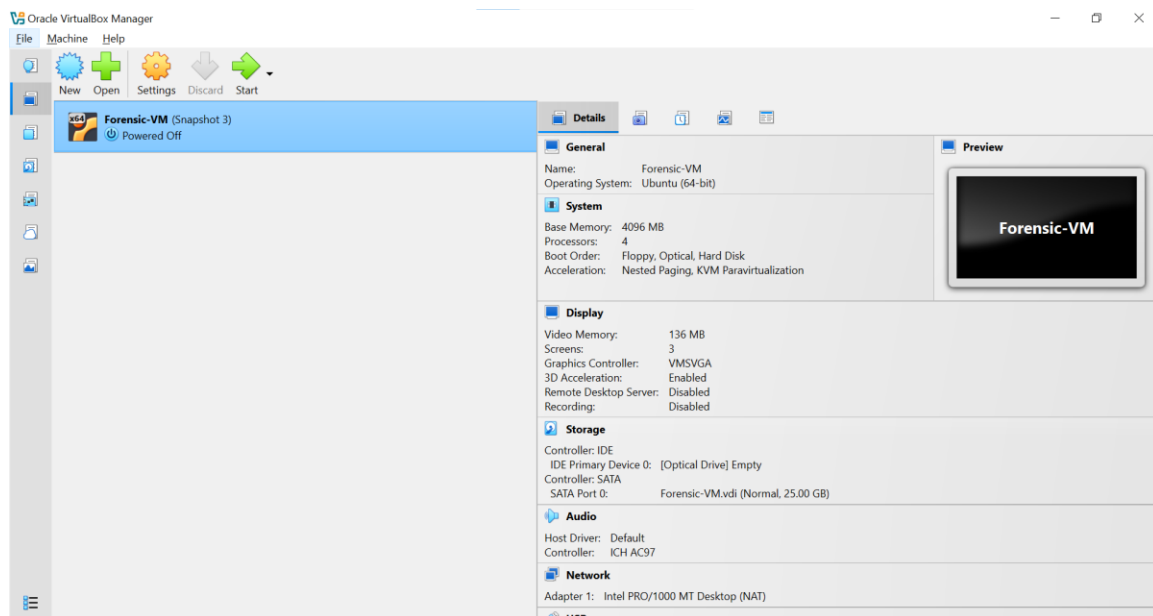
Next, a full unencrypted iTunes backup of the iPad was created on a Windows system to preserve the device's logical data in a forensically acceptable manner. An unencrypted backup was selected to maximize data visibility and ensure access to application data, file metadata, and system configuration artifacts. Care was taken to maintain data integrity throughout the acquisition process.

After acquisition, the backup data was transferred into an Ubuntu virtual machine running on Oracle VirtualBox to provide an isolated and controlled forensic analysis environment. This approach reduced the risk of data contamination and allowed repeatable analysis. Finally, the Autopsy digital forensics platform was used to examine the logical iOS backup, enabling the identification of relevant forensic artifacts such as file remnants, application usage data, and timeline information. This methodology supported the reconstruction of user activity and provided insight into how insider data leakage can be investigated using mobile forensic techniques.

Tools and Environment

Several tools were utilized throughout this investigation to support the acquisition, preservation, and analysis of digital evidence in a controlled and forensically sound manner. Apple iTunes was used to create a logical backup of the iPad, as it provides an officially supported method for extracting device data without altering the original evidence. An unencrypted backup was selected to allow access to a broader range of artifacts, including application data, file metadata, and configuration information relevant to the investigation.

Oracle VirtualBox was employed to create an isolated and controlled analysis environment by hosting an Ubuntu virtual machine. This approach ensured that forensic examination was conducted in a secure workspace, minimizing the risk of evidence contamination and enabling repeatable analysis. Autopsy served as the primary forensic analysis tool within the virtual environment. It enabled systematic examination of the extracted iTunes backup, including analysis of file structures, metadata, application-related artifacts, and timeline information. Through these tools, investigators were able to identify and correlate relevant evidence supporting the reconstruction of insider activity on the iPad.



Data Acquisition

Data acquisition is a critical phase in the digital forensic process, as it directly affects the quality and reliability of the evidence available for analysis. In this project, several sample files representing confidential corporate information were intentionally created and stored on the company-issued iPad prior to the simulated incident. These files were designed to resemble realistic business documents, such as internal reports and sensitive organizational data, to accurately simulate a corporate data environment commonly found on employee-issued mobile devices.

To ensure maximum accessibility to forensic artifacts, a full unencrypted iTunes backup of the iPad was created as part of the evidence acquisition process. Encryption was deliberately disabled to allow full visibility into application data, file metadata, and configuration files contained within the backup. This step was necessary to ensure that the Autopsy digital forensics platform could successfully parse, interpret, and analyze the backup data without restrictions imposed by encryption mechanisms. The use of an unencrypted backup enabled investigators to examine logical artifacts that are typically available in iOS backups, including remnants of corporate documents and application-related data.

The resulting iTunes backup files were stored in the default MobileSync/Backup directory on the acquisition system. Following acquisition, the backup directory was carefully transferred into a virtual machine environment for analysis. Throughout the transfer process, strict care was taken to preserve data integrity and prevent any modification or loss of evidence. This controlled handling of the backup data ensured that the forensic examination remained reliable, repeatable, and defensible, supporting accurate reconstruction of user activity during the simulated insider data-leak incident.

Backups

The screenshot displays the 'Backups' settings for an iPad. It is divided into two main sections: 'Automatically Back Up' and 'Manually Back Up and Restore'.

Automatically Back Up: This section contains two radio button options. The first is 'iCloud', with the description 'Back up the most important data on your iPad to iCloud.' The second is 'This Computer', which is selected (indicated by a blue dot) and has the description 'A full backup of your iPad will be stored on this computer.' Below these options is a checkbox for 'Encrypt local backup' with the description 'This will allow account passwords, Health, and HomeKit data to be backed up.' A 'Change Password...' button is located at the bottom of this section.

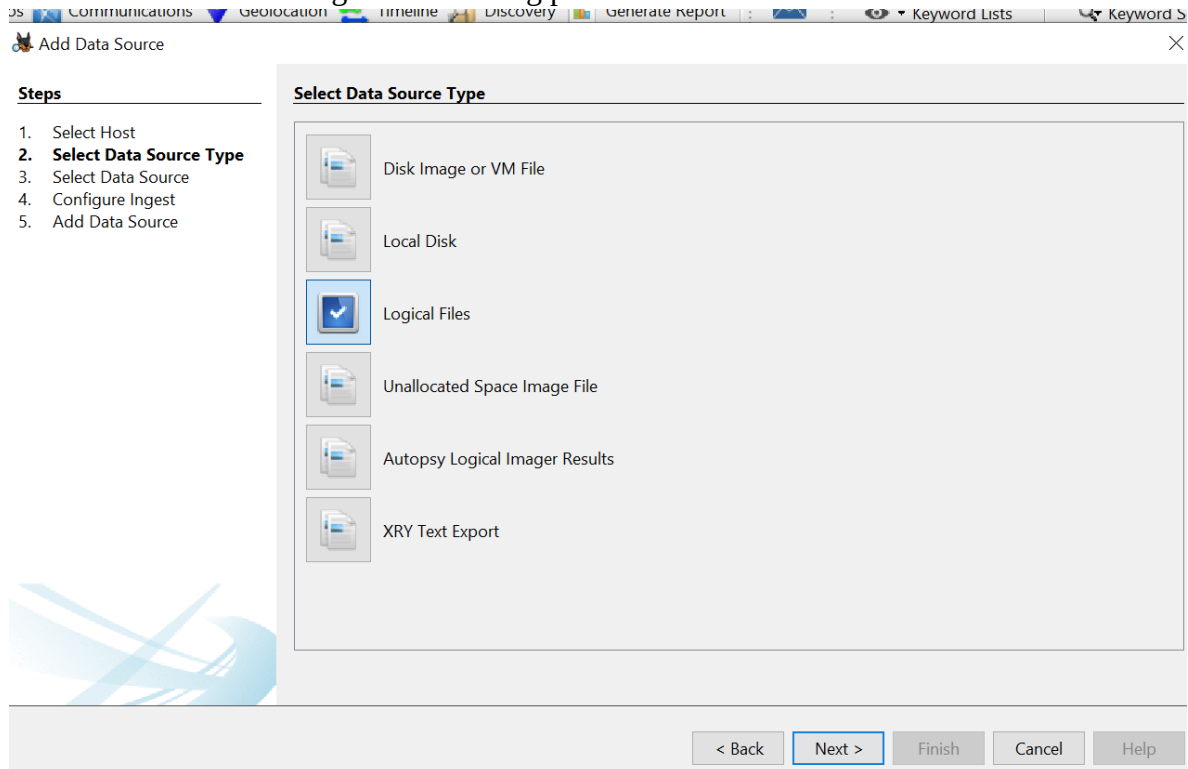
Manually Back Up and Restore: This section has the heading 'Manually Back Up and Restore' and the description 'Manually back up your iPad to this computer or restore a backup stored on this computer.' It contains two buttons: 'Back Up Now' and 'Restore Backup'.

Latest Backup: Below the manual backup section, it states 'Latest Backup: Today 4:23 PM to this computer'.

Analysis and Findings

The analysis focused on examining logical file sets, directory structures, and file metadata extracted from the iTunes backup. Using Autopsy, the investigation identified files that were created during the simulated insider activity, along with associated timestamps such as creation and modification times. These timestamps helped establish a basic sequence of user interactions with the files on the device.

In addition, artifacts related to cloud-storage applications were identified within the backup data. Although detailed upload transaction logs were not available due to iOS system restrictions, the presence of application-related files, cached data, and metadata suggests that the documents were handled in a manner consistent with unauthorized data exfiltration. When these findings were correlated with the simulated user actions, they provided circumstantial evidence supporting the conclusion that confidential files were transferred outside the organization using personal cloud services.



Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing

Metadata

Table Thumbnail Summary

16 Results

Save Table as CSV

Date Created	Organization	Description	Data Source	Program Name	User ID
2023-10-09 18:32:36 EDT	Underwriters laboratories Inc.	PowerPoint Presentation	LogicalFileSet1		
2023-10-09 18:32:00 EDT	eurecom		LogicalFileSet1	Microsoft Office Word	Tarek
2025-08-18 06:00:06 EDT		I-20 Application	LogicalFileSet1		
			LogicalFileSet1		
2024-05-30 08:07:24 EDT		Grade	LogicalFileSet1		
2018-05-12 12:08:49 EDT			LogicalFileSet1		
2025-02-06 05:13:43 EST		Copy of Black and White Minimalist Professional Resu...	LogicalFileSet1		
2023-10-18 17:31:37 EDT		Black and White Minimalist Professional Resume	LogicalFileSet1		
		Confidential	LogicalFileSet1		
			LogicalFileSet1		
			LogicalFileSet1		
2025-11-10 23:40:32 EST			LogicalFileSet1		

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

The forensic analysis focused on examining the logical file sets, directory structures, and file metadata extracted from the iTunes backup of the iPad. Using Autopsy, multiple document files created during the simulation were identified, including Word documents, PDF files, and presentation files. Each file contained associated metadata such as creation timestamps, file types, and application or organization identifiers, which provided insight into how and when the files were created or modified. In addition, metadata artifacts related to cloud-enabled document handling were observed. While direct logs of cloud uploads were not available, the presence of sensitive document files, their naming conventions, and correlated timestamps suggest that these files were actively handled in a manner consistent with potential data exfiltration. The analysis demonstrates that even in the absence of explicit upload or access logs, logical backups can still provide valuable contextual evidence that supports an insider data-leak hypothesis.

Insider Threat Mapping

The observed user behavior aligns closely with well-documented insider-threat techniques, particularly the misuse of authorized access and data exfiltration through cloud-based services. In this scenario, the user did not bypass security controls or exploit system vulnerabilities; instead, they leveraged legitimate access privileges granted by the organization to interact with confidential files. This pattern is consistent with insider threats described in frameworks such as MITRE ATT&CK, where trusted users intentionally or unintentionally abuse their access to move sensitive data outside organizational boundaries.

Although the forensic analysis could not directly capture detailed logs of file uploads, deletions, or cloud synchronization events due to limitations of iTunes backups and iOS security controls, the available metadata provides meaningful circumstantial evidence. Correlated timestamps, file naming patterns, document types, and application-related metadata collectively indicate that sensitive files were actively handled in a manner consistent with data staging and potential exfiltration. These findings demonstrate how insider activity can still be inferred through indirect artifacts, emphasizing the importance of metadata correlation in mobile forensic investigations even when explicit activity logs are unavailable.

Limitations

A significant limitation of this forensic investigation is the reliance on iTunes logical backups as the primary acquisition method. Due to Apple's strict security and privacy architecture, iOS restricts access to many system-level and application-level artifacts. As a result, several critical data elements could not be captured or examined during the analysis.

Specifically, iTunes backups do not include detailed file access logs, records of file copy or move operations, file deletion events, or cloud-upload transaction logs associated with third-party applications such as Google Drive or Dropbox. In addition, comprehensive Files application usage history and granular event timelines are stored in protected system areas that are not accessible through standard backup mechanisms. These limitations significantly reduce visibility into exact user actions and the sequence of events.

These constraints highlight the broader challenges of conducting mobile forensic investigations on iOS devices. While logical backups provide valuable contextual and metadata-based evidence, they often lack the depth required to fully reconstruct user behavior. In real-world investigations, this underscores the importance of alternative acquisition methods, such as encrypted backups with credentials, mobile device management (MDM) logs, cloud-service provider logs, or advanced forensic tools that support physical or agent-based acquisition. Understanding these limitations is essential for accurate interpretation of findings and for setting realistic expectations in insider-threat investigations involving iOS devices.

Conclusion

This project demonstrates that iOS logical backups can preserve valuable forensic artifacts that are relevant to insider data-leak investigations, even in the absence of detailed system logs. File metadata, directory structures, and application-related artifacts extracted from an iTunes backup can provide important contextual evidence regarding how sensitive files were created, modified, and handled. When analyzed using forensic tools such as Autopsy, these artifacts allow investigators to reconstruct portions of user behavior and identify patterns consistent with insider misuse.

Despite the inherent limitations imposed by Apple's security architecture, the findings show that logical backups remain a useful source of evidence for internal investigations and academic forensic analysis. However, accurate interpretation of results requires a clear understanding of both the capabilities and constraints of iOS forensic methods. By acknowledging these limitations and carefully correlating available evidence, investigators can make informed assessments while avoiding unsupported conclusions in insider-threat cases.